

Iteration For Nonlinear System (~~P-177~~)Jacobian Matrix

Assume that $f_1(x, y)$ & $f_2(x, y)$ are function of the independent variable x & y then their Jacobian matrix $J(x, y)$ is

$$J(x, y) = \begin{bmatrix} \partial f_1 / \partial x & \partial f_1 / \partial y \\ \partial f_2 / \partial x & \partial f_2 / \partial y \end{bmatrix}$$

Similarly

$$J(x, y, z) = \begin{bmatrix} \partial f_1 / \partial x & \partial f_1 / \partial y & \partial f_1 / \partial z \\ \partial f_2 / \partial x & \partial f_2 / \partial y & \partial f_2 / \partial z \\ \partial f_3 / \partial x & \partial f_3 / \partial y & \partial f_3 / \partial z \end{bmatrix}$$

Exp ~~2.30~~ 3.30 P-181

Find the Jacobian matrix $f(x, y, z)$ of order 3×3 at the point $(1, 3, 2)$ & $(3, 2, 1)$ for

$$f_1(x, y, z) = x^3 - y^2 + y - z^4 + z^2$$

$$f_2(x, y, z) = xy + yz + xz$$

$$f_3(x, y, z) = y/xz$$

SOL

$$J(x, y, z) = \begin{bmatrix} 3x^2 & -2y+1 & -4z^3+z^2 \\ y+z & x+z & y+x \\ -y/x^2z & 1/xz & -y/xz^2 \end{bmatrix}$$

$$J(1, 3, 2) = \begin{bmatrix} 3 & -5 & -28 \\ 5 & 3 & 4 \\ -3/2 & 1/2 & -3/4 \end{bmatrix}, \quad J(3, 2, 1) = \begin{bmatrix} 27 & -3 & -2 \\ 3 & 4 & 5 \\ -2/9 & 1/3 & -2/3 \end{bmatrix}$$

(16)

Seidal iteration

$$\begin{aligned} p_{k+1} &= g_1(p_k, q_k) \\ q_{k+1} &= g_2(p_{k+1}, q_k) \end{aligned} \quad \left. \vphantom{\begin{aligned} p_{k+1} &= g_1(p_k, q_k) \\ q_{k+1} &= g_2(p_{k+1}, q_k) \end{aligned}} \right\} 2D$$

$$\begin{aligned} p_{k+1} &= g_1(p_k, q_k, r_k) \\ q_{k+1} &= g_2(p_{k+1}, q_k, r_k) \\ r_{k+1} &= g_3(p_{k+1}, q_{k+1}, r_k) \end{aligned} \quad \left. \vphantom{\begin{aligned} p_{k+1} &= g_1(p_k, q_k, r_k) \\ q_{k+1} &= g_2(p_{k+1}, q_k, r_k) \\ r_{k+1} &= g_3(p_{k+1}, q_{k+1}, r_k) \end{aligned}} \right\} 3D$$

Fixed point iteration

$$\begin{aligned} p_{k+1} &= g_1(p_k, q_k) \\ q_{k+1} &= g_2(p_k, q_k) \end{aligned}$$

Convergence Near Fixed point
Theorem 2.9

$$\left| \frac{\partial g_1}{\partial x}(p, q) \right| + \left| \frac{\partial g_1}{\partial y}(p, q) \right| < 1$$

$$\left| \frac{\partial g_2}{\partial x}(p, q) \right| + \left| \frac{\partial g_2}{\partial y}(p, q) \right| < 1$$

$$\left| \frac{\partial g_1}{\partial x}(p, q) \right| + \left| \frac{\partial g_1}{\partial y}(p, q) \right| < 1$$

$$\left| \frac{\partial g_2}{\partial x}(p, q) \right| + \left| \frac{\partial g_2}{\partial y}(p, q) \right| < 1$$

Eq (17) P-183 (3D)

Ex P-190 Q5-Q8

Q5

also find $J(p_0, q_0)$

Find (a) Jacobian matrix $J(x, y)$ of partial derivative

(b) compute (p_1, q_1) & (p_2, q_2) using Fixed point

(c) = = = = = Seidal iteration

$$x = g_1(x, y) = \frac{8x - 4x^2 + y^2 + 1}{8}$$

$$y = g_2(x, y) = \frac{2x - x^2 + 4y - y^2 + 3}{4}$$

$$(p_0, q_0) = (1.1, 2.0)$$

$$(a) J(x, y) = \begin{pmatrix} \frac{\partial}{\partial x} g_1(x, y) & \frac{\partial}{\partial y} g_1(x, y) \\ \frac{\partial}{\partial x} g_2(x, y) & \frac{\partial}{\partial y} g_2(x, y) \end{pmatrix}$$

$$J(x, y) = \begin{pmatrix} 1-x & y/4 \\ \frac{1-x}{2} & \frac{2-y}{2} \end{pmatrix} \text{ Ans}$$

$$J(p_0, q_0) = J(1.1, 2) = \begin{pmatrix} 1-1.1 & 2/4 \\ \frac{1-1.1}{2} & \frac{2-2}{2} \end{pmatrix}$$

$$J(1.1, 2) = \begin{pmatrix} -0.1 & 0.5 \\ -0.05 & 0 \end{pmatrix} \text{ Ans}$$

(18)

(b) Fixed point

$$p_0, q_0 \rightarrow 1.1, 2$$

$$p_{k+1} = \frac{8p_k - 4p_k^2 + q_k^2 + 1}{8}$$

$$q_{k+1} = \frac{2p_k - p_k^2 + 4q_k - q_k^2 + 3}{4}$$

} (A)

put $k=0$ in eq (A) $p_0 = 1.1, q_0 = 2$

$$p_1 = \frac{8p_0 - 4p_0^2 + q_0^2 + 1}{8} = 1.12$$

$$q_1 = \frac{2p_0 - p_0^2 + 4q_0 - q_0^2 + 3}{4} = 1.9975$$

put $k=1$ in eq (A)

$$p_2 = \frac{8p_1 - 4p_1^2 + q_1^2 + 1}{8} = 1.116550$$

$$q_2 = \frac{2p_1 - p_1^2 + 4q_1 - q_1^2 + 3}{4} = 1.996398$$

Ans

Check convergence

$$\left| \frac{\partial}{\partial x} g_1(p, q) \right| + \left| \frac{\partial}{\partial y} g_1(p, q) \right| < 1$$

$$\left| \frac{\partial}{\partial x} g_2(p, q) \right| + \left| \frac{\partial}{\partial y} g_2(p, q) \right| < 1$$

$$\left| 1 - p \right| + \left| q/4 \right| < 1$$

$$0.11655 + 0.4990995 < 0.61564 < 1$$

$$\left| \frac{1-p}{2} \right| + \left| \frac{2-q}{2} \right| < 1$$

$$0.058275 + 0.001801 < 0.060076 < 1$$

c. Seidal iteration

$$p_{k+1} = \frac{8p_k - 4p_k^2 + q_k^2 + 1}{8}$$

$$q_{k+1} = \frac{2p_{k+1} - 2p_{k+1}^2 + 4q_k - q_k^2 + 3}{4}$$

put $k=0$ in eq (B)

$$p_1 = \frac{8p_0 - 4p_0^2 + q_0^2 + 1}{8} = 1.12$$

$$q_1 = \frac{2p_1 - 2p_1^2 + 4q_0 - q_0^2 + 3}{4} = 1.9964$$

put $k=1$ in eq (B)

$$p_2 = \frac{8p_1 - 4p_1^2 + q_1^2 + 1}{8} = 1.1160016$$

$$q_2 = \frac{2p_2 - 2p_2^2 + 4q_1 - q_1^2 + 3}{4} = 1.996632$$

Ans